When does per-turn credit assignment help?

A cross-environment study of credit assignment for multi-turn agents
(AlfWorld · SearchQA)

Agentic-RL-CA project

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Abstract

We compare four credit-assignment schemes for multi-turn RL agents — outcome-only group-relative (**GRPO**), anchor-state group-relative (**GiGPO**), and two critic-based per-turn PPO variants (**turn-PPO**, **token-PPO**) — across two environments with very different horizons: **AlfWorld** (embodied household tasks, 30–50 turns, Qwen3-1.7B) and **SearchQA** (agentic retrieval QA, a hard 4-turn cap, Qwen3-4B). The central finding is that *the ranking of credit-assignment methods reverses with horizon length.* On long-horizon AlfWorld the per-turn methods win decisively (turn-PPO 0.963, GiGPO 0.955 vs GRPO 0.881); on short-horizon SearchQA the flat outcome-only methods win (GRPO 0.441 $\approx$ GiGPO 0.446 vs turn-PPO 0.389, token-PPO 0.381). A controlled contrast isolates the mechanism: turn-level and token-level PPO are the same algorithm differing *only* in whether GAE runs over turns or tokens, and that single change is worth +0.202 on AlfWorld but only +0.008 on SearchQA — a 25× larger effect where the horizon is long. A horizon-stratified analysis explains why: per-turn credit's value scales with *genuine sequential depth.* On AlfWorld it keeps trajectories short and on-track (GRPO wastes turns wandering); on SearchQA the 4-turn cap is the binding ceiling, and the only positive horizon signal is the *critic* methods narrowing GRPO's gap on the deepest truly-sequential chains (MuSiQue, turn-PPO +0.031 $[+0.003, +0.057]$), a benefit swamped by over-search on shallow comparison questions. A thinking-mode ablation confirms the SearchQA ordering is not a truncation artifact: with `enable_thinking=False` at the same budget both arms match or beat their thinking counterparts using 66–77% fewer tokens, and GRPO still leads turn-PPO. We release all figures, the rerunnable analysis, and the recovered required-hops annotations.

1 Setup

Methods. *GRPO* — token-level, outcome-only group-relative advantage (γ=1), no critic. *GiGPO* — group-relative with an additional anchor-state step advantage (γ=0.95), no critic. *turn-PPO* — learned critic + cross-turn GAE (`gae_turn`, γ=λ=0.95). *token-PPO* — learned critic + token-level GAE (`gae`); the granularity control for turn-PPO. *floor* — the pre-RL policy.

Environments & scale (a caveat carried throughout). AlfWorld uses Qwen3-**1.7B** (reward_models campaign, single seed, RL from a replay-BC policy); SearchQA uses Qwen3-**4B** (this project, seed 0). The two are different environments and not scale-matched — cross-environment claims are about the *pattern of method rankings*, never absolute numbers. All four arms now run in both environments (the AlfWorld token-PPO leg was added 2026-07-24 to complete the granularity contrast).

Metric & protocol. AlfWorld: task success rate (seen 128-game greedy val for training curves; unseen 134-game greedy for eval). SearchQA: per-question exact match (EM) with alias normalisation over the full 7-task, 51,713-question test set (NQ, TriviaQA, PopQA, HotpotQA, 2WikiMultihopQA, MuSiQue, Bamboogle), greedy, top-3 wiki-18 retrieval, 4-turn cap. Identical harness for every checkpoint.

Required hops (H_{gold}). The SearchQA packing had stripped the datasets to question+answer; we re-attached the annotated reasoning-chain length by an exact positional join to the original FlashRAG splits (verified 51,713/51,713 question match). H_{gold}=1 for NQ/TriviaQA/PopQA, 2 for HotpotQA/Bamboogle, len(decomposition) $\in \{2, 3, 4\}$ for MuSiQue, and 2 or 4 for 2Wiki (bridge_comparison). Strata sizes: $H1$=29,190, $H2$=18,607, $H3$=760, $H4$=3,156.

2 Headline: the method ranking reverses across environments

Table 1: Evaluation accuracy. AlfWorld = unseen success rate; SearchQA = full-set macro-EM (7-task mean). Best per column green, worst trained arm red. Two things to read: (i) the *reversal* — turn-PPO beats GRPO on AlfWorld (0.963 vs 0.881) while GRPO beats turn-PPO on SearchQA (0.441 vs 0.389); (ii) *token*-PPO is worst in both environments, and catastrophically so on long-horizon AlfWorld (0.761), where it differs from turn-PPO *only* in GAE granularity. SearchQA columns are the **full 51,713-question** test set (Fig 1 plots the 2,048-question training proxy). **GRPO and GiGPO are tied**: GiGPO leads by 0.005 here and trails by 0.007 on the proxy — both below the proxy's own ∼0.01 error, so no ordering between them is claimed.

	AlfWorld	**SearchQA (per-task EM & macro)**							
Method	unseen	NQ	TriviaQA	PopQA	HotpotQA	2Wiki	MuSiQue	Bamb.	**macro**
pre-RL floor	0.440	0.292	0.517	0.346	0.271	0.331	0.074	0.360	0.313
token-PPO	0.761	0.454	0.595	0.447	0.365	0.323	0.137	0.344	0.381
turn-PPO	0.963	0.457	0.620	0.450	0.380	0.338	0.128	0.352	0.389
GRPO	0.881	0.483	0.643	0.478	0.435	0.449	0.184	0.416	0.441
GiGPO	0.955	0.490	0.647	0.469	0.447	0.399	0.187	0.480	0.446

GRPO is the *weakest* of the three long-horizon-viable arms on AlfWorld (0.881 vs 0.955/0.963) yet *among the best* on SearchQA (0.441, statistically tied with GiGPO's 0.446); turn-PPO makes exactly the opposite trip (best on AlfWorld, 0.052 behind GRPO on SearchQA). Any single-environment conclusion about credit assignment would therefore be misleading. token-PPO, meanwhile, is worst in both — but for a reason specific to granularity, which §3 isolates. The rest of the report explains the reversal.

3 Turn behaviour: RL is largely learning *how much to act*

On SearchQA, RL mostly teaches the model to use its turn budget. The pre-RL floor *under-searches* — median 2 turns / 1 search, macro-EM 0.313. RL roughly doubles search usage (GRPO/GiGPO: 3 turns / 2 searches) for a +0.13 macro lift. But more turns is not monotonically good: GiGPO *over*-searches into the cap (cap-fail 0.29/0.50/0.88/0.66 across $H1$–$H4$ vs GRPO's 0.12/0.29/0.72/0.47), which is why it fails to beat flat GRPO despite identical macro-EM. Turn-calibration, not credit fidelity, is the lever on a 4-turn budget.

On AlfWorld, the whole ranking is "how often does it wander?" Every method solves *exactly* 100% of episodes that finish within $\leq$ 20 turns (Fig 5, left; all four arms, all four sub-20 bins). The accuracy differences are entirely about *what fraction of episodes drag past 20 turns*:

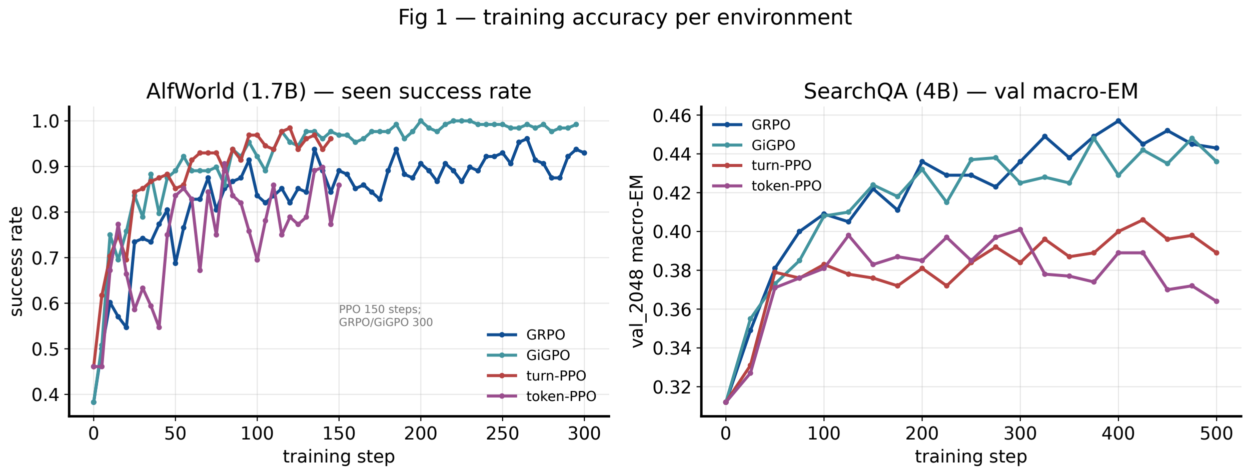

Figure 1: **Fig 1.** Training accuracy. *Left, AlfWorld:* seen 128-game success — the per-turn methods (GiGPO 0.992, turn-PPO 0.961) dominate flat GRPO (0.930). *Right, SearchQA:* **val-2048 macro-EM** — the training-time proxy on a 2,048-question subset, *not* the full-set metric of Table 1. The order inverts relative to AlfWorld: group-relative (GRPO 0.443 ≈ GiGPO 0.436) beats the critic-PPO arms (turn 0.389, token 0.364). **NB** GRPO and GiGPO swap places between this proxy (−0.007) and the full-set eval (+0.005); the proxy tracks GRPO to 0.002 but under-reads GiGPO by 0.010, so the two are best read as *tied* (see Table 1 note). (AlfWorld PPO trained 150 steps vs 300 for GRPO/GiGPO.)

	GiGPO	turn-PPO	GRPO	token-PPO
episodes >20 turns	4.5%	11%	21%	46%
of which succeed	0.00(n=6)	0.67	0.43	0.48
mean turns	9.1	10.9	14.9	24.2
unseen success	0.955	0.963	0.881	0.761

A concrete GRPO failure mode, *"examine the book with the desklamp"* solved in 44 turns:

`t0 go to shelf 1` → `t1 go to shelf 2` → `t3 go to bed 1` → `t4 take book 1` → `t5 go to desk 1` → `t6..t13 look / look / look / look / look / look / look / look`

GRPO loops the no-op `look` eight times; GiGPO solves the same task type in **3** turns. Per-turn credit penalises the wasted steps directly; the outcome-only signal does not.

The cleanest test: turn-level vs token-level GAE

turn-PPO and token-PPO are the *same algorithm* — same learned critic, same BC initialisation, same γ=λ=0.95, same 150-step budget, same eval checkpoint. They differ in exactly one thing: whether the GAE recursion runs over **turns** or over **tokens**. That single change is worth:

	turn-level	token-level	Δ
AlfWorld (long horizon, ≤50 turns)	0.963	0.761	+0.202
SearchQA (short horizon, 4-turn cap)	0.389	0.381	+0.008

A **25× larger effect on the long-horizon environment**, from an identical algorithm differing only in credit granularity. The mechanism is visible in the discount: at γ=0.95 per *token*, an

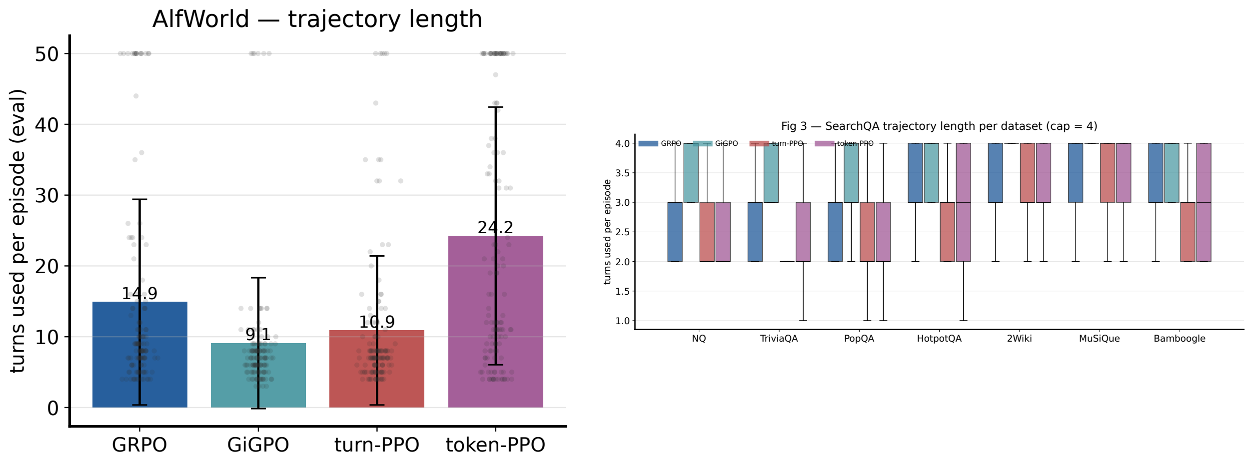

Figure 2: **Fig 2 (left):** AlfWorld trajectory length — token-PPO uses **24.2** turns on average and GRPO 14.9, vs GiGPO 9.1 / turn-PPO 10.9. **Fig 3 (right):** SearchQA turns per dataset (cap 4); turn usage rises with task difficulty and the trained arms sit well above the under-searching floor.

AlfWorld transcript of ~25k generated tokens discounts early credit by $0.95^{25000} \approx 0$, so the value signal cannot reach the decisions that matter, and the policy wanders (46% of episodes exceed 20 turns; `cool` collapses to 0.29 success at 43.6 turns). Under a 4-turn SearchQA cap the same discount spans only a few hundred tokens, so granularity is nearly irrelevant — and the two arms tie. This is the sharpest single piece of evidence for the paper's thesis, because it holds everything except granularity fixed.

Robustness: does the SearchQA ranking survive without thinking mode?

The SearchQA results above use Qwen3's thinking mode with a 2048-token response budget, under which the critic arm clips 8.4% of its turns — a clipped turn emits no action and so wastes one of only four. That makes truncation a live confound for the report's central SearchQA claim (flat GRPO > critic turn-PPO): the gap might be a truncation artifact rather than a credit-assignment one. We therefore re-ran both arms with `enable_thinking=False` at the *same* 2048 budget, 500 steps, batch, seeds and γ=λ=1 — so *only* the thinking flag differs.

Table 2: Thinking-mode ablation, SearchQA 4B (val-2048 macro-EM at step 500; full-set evals in progress). Non-thinking matches or beats thinking at ≈4× fewer tokens per turn, and the GRPO-over-turn-PPO ordering **survives** — the gap narrows from 0.054 to 0.030 but does not close.

	macro-EM			**mean resp. tokens/turn**	
	thinking	non-thinking	Δ	thinking	non-thinking
GRPO	0.443	0.452	+0.009	476	111
turn-PPO	0.389	0.422	+0.033	610	207
GRPO − turn-PPO	0.054	0.030			

Three readings. (i) **The finding is robust**: GRPO still leads turn-PPO when the truncation channel is largely removed (turn-PPO's clip falls 0.084 → 0.057), so the ordering is not a clipping artifact — though the narrowing from 0.054 to 0.030 says clipping was contributing *part* of it. (ii) **Thinking is not paying for itself here**: both arms do at least as well without it while generating 66–77% fewer tokens, i.e. the 4-turn budget rewards deciding *what to search for* far more than deliberating about it. (iii) **The critic arm gains most** (+0.033 vs +0.009), consistent with it having been the arm penalised by truncation. *Caveat:* single seed per cell, and GRPO's +0.009 sits inside that run's own step-to-step oscillation band (0.429–0.457 over steps 250–400); the defensible claim is that non-thinking is *not worse and much cheaper*, not that it is better.

4 Horizon-stratified analysis

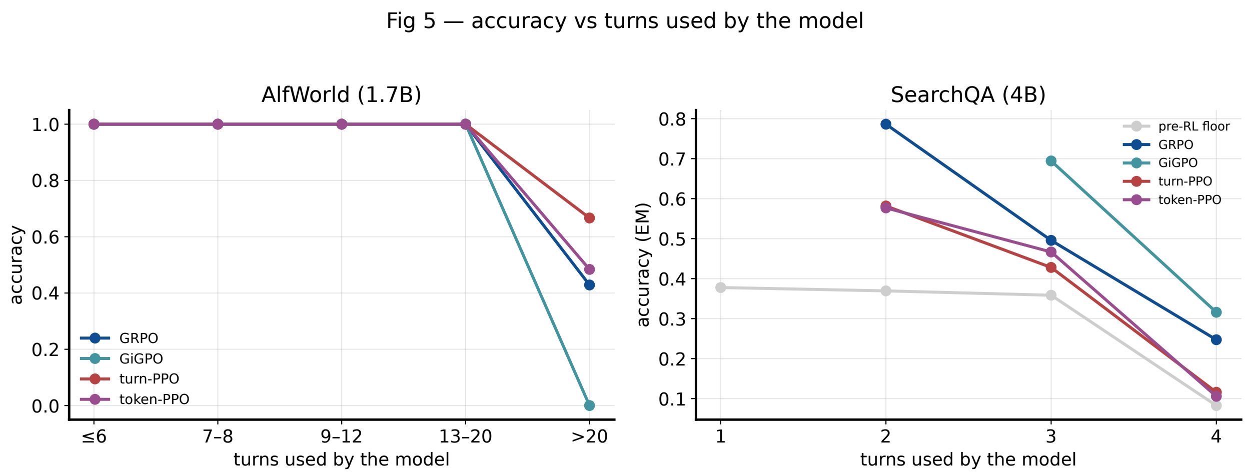

Figure 3: **Fig 5.** Accuracy vs turns *used*. *Left, AlfWorld:* all methods succeed until > 20 turns, where the wandering tail (largest for GRPO) fails. *Right, SearchQA:* accuracy falls with turns used — long trajectories are the hard, cap-hitting ones — and the critic-PPO arms sit below GRPO/GiGPO at every turn count.

EM vs required hops (within task). MuSiQue collapses monotonically for every method. On 2Wiki the floor and GRPO *rise* at the 4-hop (bridge_comparison) stratum — the comparison/base-rate shortcut — while the over-searching arms fall, and the critic arms end *below the untrained floor* there (0.224/0.138 vs 0.456): learned search displaces the floor's successful parametric early answers. The MuSiQue–2Wiki gap is the crux of §5.

Table 3: Within-task EM by H_{gold}.

	MuSiQue			2Wiki	
	H2	H3	H4	H2	H4
floor	.110	.042	.025	.297	.456
GRPO	.268	.114	.054	.433	.504
GiGPO	.276	.114	.047	.406	.372
turn-PPO	.197	.064	.035	.370	.224
token-PPO	.217	.063	.025	.374	.138

Horizon-gain vs GRPO (A2). We measure, for each arm m, the top-minus-bottom contrast of $\Delta_m(h)=\mathrm{EM}_m(h)-\mathrm{EM}_{\mathrm{GRPO}}(h)$ with a 1000-resample item bootstrap; a horizon effect is claimed only when the CI excludes 0.

Reading: turn-PPO is worse than GRPO at *every* MuSiQue hop, but its gap *narrows* as hops deepen (Δ: $-0.071 \rightarrow -0.019$) — a small, significant positive contrast. The critic's per-turn value estimates do buy something on genuinely-sequential chains; it is simply overwhelmed by over-search losses elsewhere and by lower overall performance. GiGPO's group-relative step advantage shows no such MuSiQue signal.

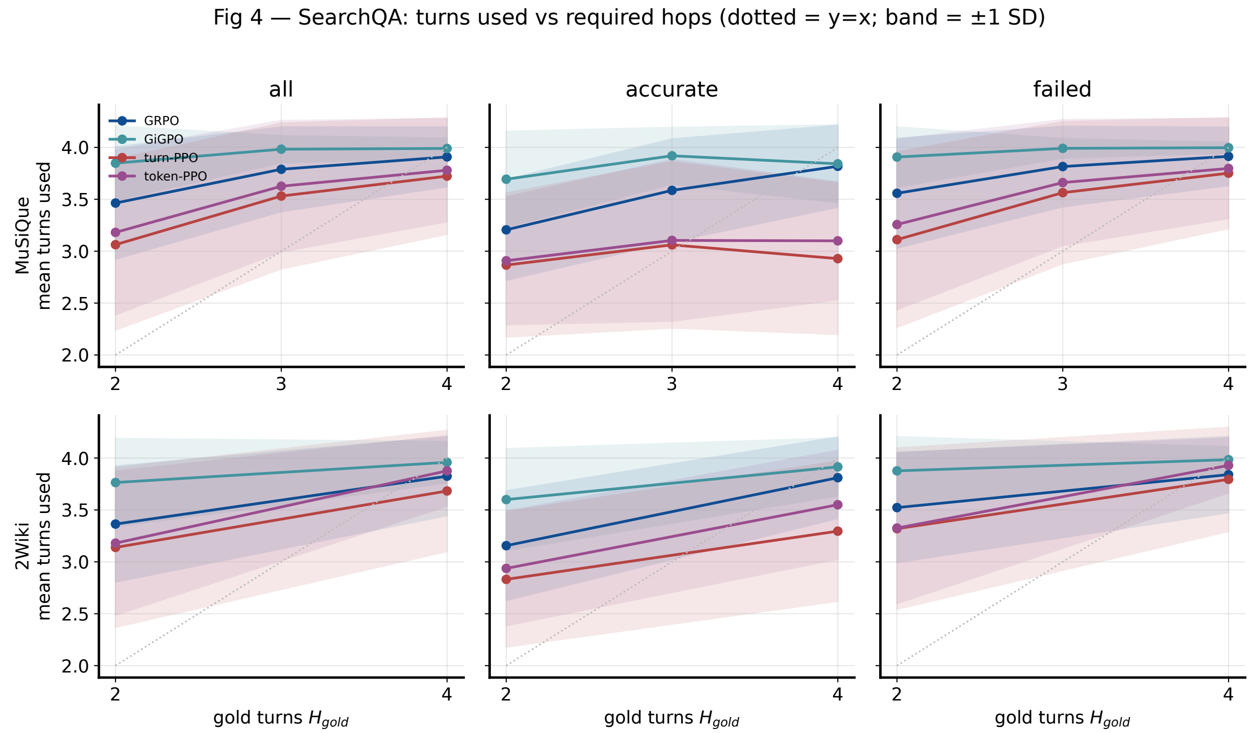

Figure 4: **Fig 4.** SearchQA turns used vs required hops, split accurate/failed/all (band ±1SD; dotted $y{=}x$). All methods escalate turns with hops on MuSiQue and pin at the cap by $H{=}3$; failed trajectories sit at the cap (budget exhausted) while successes finish earlier.

Table 4: Horizon-gain contrasts vs GRPO (Δ(high)$-\Delta$(low), 95% CI). * = CI excludes 0. The *only* positive signal is the critic methods on MuSiQue's deep chains; every arm loses on 2Wiki by over-searching shallow comparison questions.

Method	within-MuSiQue	within-2Wiki	pooled
turn-PPO	$+0.031$ [$+0.003, +0.057$]*	$-0.217^\star$	$-0.183^\star$
token-PPO	$+0.007$ [$-0.018, +0.032$]	$-0.307^\star$	$-0.233^\star$
GiGPO	-0.011 [$-0.035, +0.013$]	$-0.106^\star$	$-0.092^\star$

5 Why hop-depth crushes MuSiQue but not 2Wiki

MuSiQue's hop count is genuine *sequential* depth; 2Wiki's "4-hop" (bridge_comparison) is a *shallow parallel comparison* the label over-counts. A MuSiQue 4-hop is a chain in which each hop's answer feeds the next (the annotation uses `#1/#2/#3` placeholders):

> *"president of the newly-independent country that established the Timor-Leste Commission …with the country containing the airport that includes Lion Air?"*
> airport of Lion Air → `Juanda` → its country → `Indonesia` → +Timor-Leste Commission → `East Timor` → its president → `Francisco Guterres`.

You cannot begin hop 4 until hops 1–3 resolve. With a 4-turn budget (≈3 searches), the model runs out mid-chain — here GRPO is still decomposing at the cap and scores 0:

```
turns=4, em=0: ``<think> …the length of the city where the Yongle Emperor greeted a person the
Ming court thought were representatives sent by …Yaxing Coach's headquarters. First, I need to
break do[wn]…'' [budget exhausted]
```

A 2Wiki bridge_comparison, by contrast, is two *independent* 2-hop facts joined by a comparison (*"which film's director was born later, A or B?"*): the branches don't depend on each other (depth 2), the answer is one of two named entities (∼50% base rate), and parametric knowledge often supplies one side. Hence (i) MuSiQue is *cap-limited* — cap-fail 0.72 at $H3$, and no credit scheme overcomes "4 turns can't do 4 hops"; (ii) 2Wiki-4hop is *easier* than 2Wiki-2hop for every method, floor included, so the difficulty structure is a property of the tasks, not of the credit signal. This is the composition confound that makes pooled cross-stratum curves misleading and the within-task curves the honest ones.

6 Per-method verdict

- **GRPO** — best flat baseline. Tied for best on SearchQA with GiGPO (cheap, no critic, well-calibrated turn use; 0.441 vs 0.446 full-set, ordering reverses on the proxy) but wastes the loose AlfWorld budget wandering (14.9 turns; `look`-loops), making it the *worst* long-horizon arm.
- **GiGPO** — best long-horizon efficiency (AlfWorld 0.955 at 9.1 turns). On SearchQA it *ties* GRPO on EM but over-searches into the cap and gains nothing on the hop axis — its edge is variance/stability, not credit fidelity.
- **turn-PPO** — best on AlfWorld (0.963). On SearchQA it underperforms overall yet is the *only* arm with a significant positive horizon-gain, specifically on MuSiQue's deep chains — the clearest evidence that critic-based per-turn credit helps in proportion to sequential depth.
- **token-PPO** — worst arm in *both* environments, and the informative failure. On SearchQA (0.381) it merely trails, with the same MuSiQue-leaning / 2Wiki-losing shape as turn-PPO. On AlfWorld it collapses to 0.761 at 24.2 turns/episode — 46% of episodes exceed 20 turns — because per-token discounting annihilates credit over a 25k-token transcript. Identical to turn-PPO in every respect but granularity, it is the control that turns the paper's thesis from correlation into a controlled contrast.

7 Discussion & limitations

Thesis. The value of fine-grained per-turn credit assignment scales with *genuine sequential horizon*. Long, truly-sequential tasks (AlfWorld) reward it with efficient, on-track trajectories; short or shallow tasks (most of SearchQA, under a 4-turn cap) do not, and flat outcome-only GRPO is competitive and cheaper. Where SearchQA *is* deep (MuSiQue 3–4 hop), the critic methods leave the faintest positive trace — but the 4-turn cap, not the credit signal, is the binding constraint (the actionable fix there is more turns, not better credit).

Limitations. (1) Mixed scale (AlfWorld 1.7B, SearchQA 4B) — we compare *patterns*, not levels. (2) AlfWorld is single-seed, with unequal training budgets (both PPO arms 150 steps vs 300 for GRPO/GiGPO); the turn-vs-token contrast is budget-matched at 150, but the PPO-vs-GRPO/GiGPO comparisons are not. (3) H_{gold} is annotated chain length, ordinal, not "expected searches"; the 3-hop stratum is MuSiQue-only. (4) SearchQA metric is greedy val-protocol EM on the full test set. (5) Fig 5's accuracy-vs-turns axis is partly mechanical (hard items take more turns).

Assets, rerunnable analysis, and H_{gold} table: `docs/reports/2026-07-21_4b_horizon_eval/` (see `README.md`).